Comprehensive Exam

Toward a Comprehensive Understanding of Qualitative Human Factors Studies in Cybersecurity

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Abstract

In cybersecurity, the human-in-the-loop approach emphasizes the importance of understanding human behavior and its influence on security outcomes. Numerous works have explored this perspective, highlighting the critical role that human factors play in cybersecurity. This write-up reviews these studies to examine how researchers have approached the human aspects of cybersecurity by closely analyzing their study designs, goals, and methodologies. By synthesizing insights from the literature, this review aims to clarify the contributions of qualitative research methods in the field, showcasing their value in informing and enhancing cybersecurity practices. Ultimately, it underscores the necessity of integrating human factors into cybersecurity research to advance the Science of Security.

1 Introduction

Despite technological advancements, cybersecurity threat landscape continues to grow with alarming regularity [37]. Prior work points out that attackers have paid more attention to the human element in security—*human-in-the-loop*— than security designers have, and they have managed to exploit this advantage prodigiously [54], for instance, to acquire passwords by employing social engineering techniques [5]. Social engineering tactics—such as robocalls [75]—may manipulate users into divulging confidential information, further exacerbating the risks [45]. Additionally, cybersecurity threats such as phishing attacks can lead to identity theft when users unknowingly provide sensitive information through fraudulent emails [18, 20]. Ransomware is another growing threat that encrypts a user's files and holds the decryption key until a ransom is paid by the victim [68].

Given the persistence of cyber threats, technical solutions such as intrusion detection systems, encryption, antivirus software, and firewalls are essential, but they are not sufficient on their own. This raises a fundamental question: *Why do purely technical solutions fail to protect us effectively?* One possible explanation is that, while it is well-known that attackers exploit the human-in-the-loop vulnerability and that humans are invariably described as the "weakest link" in the security chain [84], the human factor in security has been neglected by security developers and researchers until recently [67].

The concepts of usable security and user-centered design emerged to address the gap between security and usability. In the late 1990s, three classic academic works put security usability in spotlight and paved the path for the now-popular field of `Usable Security and Privacy`. Zurko and Simon introduced the phrase *user-centered security*, identifying the crucial role of user perceptions, characteristics, needs, and abilities to have effective and robust system design [91]. Adams and Sasse reclassified users as potential parts of the solution, not the problem [5]. Meanwhile, Whitten and Tygar highlighted that even technically proficient users struggled to encrypt and decrypt emails using PGP 5.0 due to a mismatch between their task perceptions and the PGP interface, ultimately leading to recommendations for redesigning the interface to enhance usability [84].

Human behavior and decision-making play a crucial role in cybersecurity [89], influencing various aspects such as password strength and compliance with security policies—where human practices can either reinforce or weaken security measures. Prior work demonstrates that human behaviors can significantly impact cybersecurity, as user-centered security design encourages users to adopt and adhere to effective security measures [91], while failing to consider user perspectives can lead to non-compliance [5]. Furthermore, a mismatch between user perception and interface design can cause errors that compromise security [84], and a lack of motivation and understanding of security policies can result in negligence [67].

In essence, cybersecurity should be viewed as a socio-technical system where human behaviors, perceptions, and organizational practices are considered, explored, and understood. Recognizing the crucial role of users in the security landscape is essential for developing effective strategies to mitigate cyber threats and create more resilient cybersecurity systems.

2 Background

Human factors play a crucial role in cybersecurity, as users' interactions with secure systems can significantly influence security outcomes. However, secure systems have traditionally shown a remarkable indifference to user needs [91]. Ignoring these aspects often leads to systems that are prone to failure, as users are forced to navigate interfaces that don't align with their abilities and expectations.

Passwords continue to be the dominant method for access control, yet they present a persistent challenge in cybersecurity [35]. Security guidelines advocate for unique, complex passwords; however, users often choose easily memorable ones or reuse them across platforms. When users struggle with such requirements, they may resort to insecure workarounds, like writing passwords down or using simple patterns [43, 66]. This tendency can create vulnerabilities [5]. Password managers offer one remedy by simplifying password management [46], but understanding why users engage in insecure practices is key. For example, users' cognitive and behavioral motivations behind these choices—password creation, selection, and reuse practices [86, 77, 22]—may allow researchers and professionals to develop more intuitive and effective security strategies and tools [11].

2.1 Significance of human-in-the-loop

Secure system designers often aim to minimize human involvement in security tasks, as it is widely recognized that humans can be unreliable in security roles. Nonetheless, when fully automated alternatives—such as automatic system updates—are unavailable or impractical, secure systems frequently depend on a "human in the loop" to perform critical security functions [14].

However, user perspectives on security are complex and can be broadly categorized into two views: a) *Irrational*, where users are perceived as lazy or unmotivated to prioritize security (often doing the bare minimum), and b) *Rational*, where users reject security advice based on an economic calculus, assessing the trade-offs between effort and security benefits [33]. In practice, users often evaluate security measures by weighing the inconvenience or effort involved against their personal perception of risk. If they perceive the likelihood or impact of a security threat as low, they may prioritize convenience, choosing to bypass or ignore security recommendations [33]. In a recent study, Redmiles et al. [60] found that users' past behavior, knowledge of costs, and awareness of risks and context influence their security decision-making. For example, many people avoid software updates that could disrupt their workflow, even if these updates address critical security vulnerabilities [79]. This trend reflects a broader challenge in cybersecurity: finding a balance between user convenience and stringent security requirements.

Overall, the notion that users are the "weakest link" in cybersecurity is both outdated and counterproductive. Rather than viewing users as obstacles, we should design systems that leverage their strengths and address their weaknesses, transforming them into active participants in the security process. By considering users' natural working patterns and behaviors, we can create more intuitive and effective security measures that enhance overall system security.

2.2 To understand users and build systems for ALL

In psychology, mental models are a tool for understanding how individuals perceive and reason about various situations. These models represent a person's internal idea of how something functions—whether it's a real-world system, a hypothetical scenario, or even an imaginary concept [30]. Mental models even guide decision-making by allowing people to mentally simulate the outcomes of their actions [28].

Moreover, users' decision-making [23], perceptions of risk [76, 24], and security and privacy practices [48, 36] vary widely and are often influenced by factors such as convenience, personal experience, and perceived threats. For example, systems designed with simplicity in mind, as discussed by [16], are more likely to be effectively used by people, thereby enhancing security. Given that user experiences and decision-making skills vary widely, Redmiles et al. [57] conducted an online survey with 526 U.S.-census representative users in 2016 to streamline the learning of key security practices. Their findings revealed evidence of a "digital divide" that may disproportionately affect disadvantaged users.

Prior research has shown that users frequently find security protocols across various areas—such as online security [39, 82], Wi-Fi security [41], and social media sharing [38]—too complex to understand fully. As a result, they may adopt coping strategies that lead to sub-optimal security outcomes, such as bypassing security features altogether [50, 16]. Consequently, efforts to improve the usability of privacy and security technologies have become increasingly important [15], such as SSL warnings [8]. A solution to that is user education and training.

Research suggests that good security advice should be practical, actionable, and easy to comprehend [61, 63]. However, online security recommendations often fail to be actionable for several reasons: the cost-benefit ratio may not justify the effort [33, 23], there may be an overwhelming amount of advice without clear prioritization [61, 53], or the effectiveness of specific advice may fluctuate in response to evolving threats, emerging technologies, and user experiences [63]. Furthermore, a disconnect often exists between the perceptions of security experts and non-experts regarding what constitutes "good" advice [36].

Ultimately, human factors can directly impact the efficacy of security measures, making it essential to consider user insights when designing and implementing security protocols as well as tailoring security advice that resonates with their perceptions and decision-making processes.

3 Approach for Literature Review

To design effective security systems, it is essential to understand how humans interact with them. To contribute to the field of Usable Security and Privacy, a systematic literature review was conducted to gather a comprehensive understanding of the field, targeting well-regarded venues such as USENIX, IEEE Security and Privacy (S&P), SOUPS, ACM CCS, and CHI.

Academic databases were utilized to identify relevant papers using keywords (human factors, cybersecurity, usable security, privacy, interviews, surveys, user study, etc.) related to usable security and privacy. The search process included not only the initial identification of papers but also the exploration of related articles through citation tracking and references. This approach allowed for a more thorough examination of the existing literature, ensuring that significant studies relevant to human factors in usable security and privacy were included in the review.

Papers were included based on their relevance to human factors studies, prioritizing qualitative research to capture user experiences, behaviors, and perceptions, i.e. *papers that investigate human factors to understand what makes things go the way they go.* To begin exploring the question, "What constitutes human factors research in cybersecurity?" the collected papers were analyzed to find patterns in their overall study approaches.

4 Methodologies in Human Factors Research in Cybersecurity

A systematic review of literature across key venues revealed common methodologies that characterize human factors research in this field. The studies reviewed reflect a blend of observational, qualitative, and evaluative approaches, categorized as follows:

- Observe: User interaction observations
- Inquire: Interviews and surveys gathering qualitative data
- Evaluate: Assessments of security interventions' effectiveness

This categorization offers a structured perspective on how research in usable security and privacy is conducted. It not only clarifies the focus and methods of each paper but also helps identify patterns in research approaches, thereby offering a comprehensive view of how user experiences are studied in the field.

Details and examples about each type are provided next.

4.1 Observation

Observation type works involved studying how users interact with security systems (in real-time), identifying pain points, and behaviors that lead to security lapses.

Contextual Inquiry of Android Permissions [85]. This study involved instrumenting the Android platform to gather data on how frequently and under what conditions smartphone applications access protected resources requiring permissions. Conducted with 36 participants, the research aimed to explore "contextual integrity," focusing on unexpected application access. The analysis included 27 million data points and exit interviews, revealing that over 80% of participants wished to deny at least one permission request, expressing a desire to block more than a third of all requests. These findings suggest the potential for future systems to automatically identify situations where users prefer to make security decisions regarding permission access.

Impact of Password Managers on Password Strength and Reuse [46]. The study investigates the influence of password managers on users' password strength and reuse, addressing the ongoing security issues associated with passwords as an authentication method. Researchers combined qualitative data from an online survey of 476 participants, focusing on password creation and management strategies, with quantitative data collected from 170 users via a browser plugin that tracked password metrics and entry methods. The findings indicate that password managers can affect password security positively or negatively, depending on users' strategies and the effectiveness of the managers in supporting proper password management from the outset. The study suggests further research on enhancing password managers to better align with user needs and improve overall password security.

An Experience Sampling Study of User Reactions to Browser Warnings [62]. This study investigates user decision-making regarding web browser security warnings, focusing on comprehension and adherence rates. By conducting an experience sampling study with over 6,000 Chrome and Firefox users, researchers aimed to understand the reasons behind users' actions when confronted with real warnings. The findings reveal that users' reasons for adhering to or ignoring warnings are varied and context-dependent. Unlike previous research, the study did not identify a single dominant failure in modern warning design, such as habituation. Instead, it concludes that improving adherence rates will require addressing a range of smaller contextual misunderstandings about the warnings.

What, Why, and How of Hacker Mistakes [27]. The study examines mistakes made by security content creators during vulnerability exploitation in Capture The Flag (CTF) challenges by analyzing 30 YouTube screencasts from 11 hackers. Researchers identified 124 distinct issues, investigating their types, underlying causes, and the time spent on each. The study highlights the cognitive and behavioral factors contributing to these mistakes and aims to enhance education programs and automated tools to improve vulnerability research efficiency. Overall, it emphasizes the need to understand these errors to mitigate their impact on security practices.

4.2 Inquiry

Inquiry type studies typically included interviews [82, 39, 1, 59, 83, 10, 55, 48], focus groups [70], and surveys [31, 49, 73] to gather users' perceptions, experiences, and attitudes towards security measures.

Understanding User Motivations for Smartphone Locking Behaviors [21] This study explores the motivations behind smartphone users' decisions to lock or not lock their devices. Despite over 90% of Americans owning mobile phones that store sensitive data, 35% do not use locks due to concerns about emergency identification and a belief that their data is not valuable. Interviews with 28 participants revealed that 29% did not lock their devices, a finding supported by a survey of 2,518 users. An experiment with 995 participants indicated that many had sensitive information stored in their email accounts, contradicting their perceptions of data security. The research emphasizes the need for design improvements in mobile security to address user concerns and enhance protection.

A Comparison of Software Vulnerability Discovery Processes [80]. This paper investigates the processes and interactions between software testers and white-hat hackers in identifying security vulnerabilities. Conducting semi-structured interviews with 25 participants, the study reveals that while both groups follow similar vulnerability discovery processes, they achieve different outcomes due to varying experiences and knowledge of security concepts. The findings highlight the need for improved security training for testers, better communication between hackers and developers, and more effective bug bounty policies to enhance hacker participation. Ultimately, the research aims to contribute to a more secure software development ecosystem by fostering collaboration between testers and hackers.

Comparing Expert and Non-Expert Security Practices [36]. The paper addresses the inadequacy of current online safety advice, which often imposes unrealistic or overly time-consuming expectations on users. To enhance security recommendations, the study seeks to identify the online safety practices that individuals consider most important. It compares the self-reported security behaviors of non-experts with those of security experts, defined as individuals with at least five years of experience in computer security. The study involved two online surveys, one with 231 security experts and another with 294 participants from Amazon Turk (MTurk). Results reveal a significant gap between the security practices of experts and non-experts; experts tend to prioritize actions like installing software updates, using two-factor authentication, and employing password managers, while non-experts favor antivirus software, visiting familiar websites, and frequently changing passwords.

A Qualitative Study on Security Operations Center Issues [42]. This study examines the challenges faced by Security Operations Centers (SOCs), which are crucial for managing security incidents in organizations. Despite their importance, there is limited research on SOC-specific issues. To address this gap, the authors conducted 18 semi-structured interviews with SOC analysts and managers across various industries. The analysis revealed both technical and non-technical problems within SOCs, including significant disagreements between managers and analysts that could threaten operational efficiency and effectiveness. The study concludes by offering insights for both academic research and SOC management, emphasizing the need for improvements in SOC operations.

4.3 Evaluation

Evaluation type studies involved testing the effectiveness of security measures through usability studies [11, 2], surveys [12], in-the-wild/lab/online (between-subjects) experiments [26, 18, 78], and feedback processes [17].

Studying GDPR Consent Notices in the Field [78]. This study explores user fatigue with cookie consent notices following the GDPR's implementation and examines how interface design influences user interactions. Conducting experiments with over $80,000$ unique users on a German website, the authors find that notices in the lower left of the screen yield higher engagement, and users are more likely to accept tracking when given binary choices. The results highlight the importance of regulatory guidance not only to require consent but also to specify effective design practices that facilitate informed user choices.

The Impact of Information Sources on Code Security [3]. This paper investigates the relationship between the use of information resources and code security among Android developers. Through a survey of 295 app developers and a subsequent lab study with 54 participants, the research finds that those using Stack Overflow produced significantly less secure code compared to those using official Android documentation or books. While Stack Overflow was found to be more accessible, it often led to insecure coding practices, highlighting the necessity for secure yet user-friendly documentation to support developers in creating safer applications.

Exploring Adoption of Two-Factor Authentication at a University [12]. This study evaluates the adoption of two-factor authentication (2FA) at Carnegie Mellon University (CMU) to understand user behaviors and barriers related to its implementation. Through observation surrounding a mandatory adoption deadline, the researchers found that while adopters found 2FA somewhat annoying, they considered it easy to use and believed it enhanced account security. Positive experiences with CMU Duo often encouraged users to adopt 2FA for other accounts. The study also highlights

misconceptions, insecure practices, and design issues among users, leading to recommendations for optimizing large-scale 2FA deployments through effective design, mandates, and messaging strategies.

User Attention, Comprehension, and Behavior toward Android Permissions [25]. This study evaluates the effectiveness of Android's permission system in informing users about the risks associated with application installation. Through an Internet survey of 308 users and a laboratory study involving interviews with 25 users, the researchers found that user attention and comprehension regarding permission requests were alarmingly low; only 17% of participants paid attention to permission warnings, and just 3% could correctly answer all comprehension questions related to those permissions. These findings suggest that the current permission warnings fail to aid most users in making informed security decisions. However, a small group of users demonstrated awareness and understanding, prompting the researchers to offer recommendations for enhancing user attention and comprehension regarding permissions, as well as highlighting ongoing challenges in this area.

5 Conducting Qualitative Human Factors Research

5.1 Study Protocol

Developing the study protocol typically involves creating a structured yet flexible interview guide or survey that addresses key research questions. For example, questions are formulated to probe participants' experiences [90], motivations [20], and decision-making processes [88] in the context of cybersecurity. Pilot studies are often conducted with a small subset of participants to refine the protocol, ensuring clarity, relevance, and efficiency in data collection [83].

5.2 Data Collection Methods

The studies reviewed utilize a variety of data collection methods tailored to their specific research goals and methodologies. These methods can be broadly categorized into qualitative and quantitative approaches, each serving distinct purposes in understanding user behaviors and perceptions related to cybersecurity. Qualitative methods provide depth and context, uncovering the reasons behind user behavior. Quantitative methods, on the other hand, offer breadth and generalizability, revealing patterns and trends across larger populations.

5.2.1 Qualitative Data Collection

Qualitative data collection is prevalent in both Inquiry (§ 4.2) and Observation (§ 4.1) studies, providing in-depth insights into user experiences and motivations. Common methods include:

Interviews: Semi-structured interviews allow researchers to explore participants' thoughts and feelings in detail. Semi-structured interviews employ a blend of closed- and open-ended questions, often accompanied by follow-up why or how questions [6].

This method facilitates an open dialogue, enabling participants to express their perspectives on security measures and their experiences with security systems. For example, in a 2010 study, Rick Wash conducted two rounds of semi-structured interviews to understand how home computer users understand and make of security threats [82].

Haney et al. used semi-structured interviews to understand how cybersecurity advocates motivate their audiences to engage in beneficial security behaviors [32]. They reported:

> *We chose semi-structured interviews over other methods, such as surveys, because of the richness of data afforded, the latitude to ask follow-up questions to clarify or delve deeper into participant responses, and the ability to encourage participants to add other relevant information not explicitly targeted [13].*

Focus Groups: This method involves discussions among groups of participants to gather diverse views on a topic. It can reveal social dynamics and shared beliefs that influence user behavior. For instance, semi-structured interviews and focus groups in Simko et al. offered rich discussions about refugees use of and barriers to technology and their security & privacy practices [70] . They reported their choice and approach for using focus groups:

> *1) We conducted interviews until reaching thematic saturation, and then turned to focus groups of refugees.*
> *2) Our use of focus groups, rather than one-on-one interviews, enabled free-flowing conversations with the refugees, and in less intimidating settings than one-on-one interviews.*

5.2.2 Quantitative Data Collection

Quantitative data collection methods are often employed in both Inquiry (§ 4.2) and Evaluation (§ 4.3) studies to assess the effectiveness of security measures. Common techniques include:

Surveys: Surveys are widely used to collect structured data from a large number of participants. They often include Likert-scale questions to measure attitudes toward security practices and behaviors. For instance, surveys in studies like Colnago et al. assessed user perceptions and experiences with two-factor authentication [12]. Redmiles et al. has summarized best practices for conducting surveys in security and privacy research [56].

Experiments: Controlled experiments, either in the lab or in the field, are used to measure specific outcomes—such as usability or effectiveness—related to security interventions. For example, the study by Utz et al. investigated user interactions with GDPR consent notices through an experimental design and follow-up surveys, allowing for quantitative assessment of user engagement [78].

5.2.3 Mixed-Methods Approaches

Some studies adopt a mixed-methods approach, integrating qualitative and quantitative data to provide a more comprehensive understanding of human factors in cybersecurity. This method allows researchers to triangulate findings, combining the depth of qualitative insights with the breadth of quantitative analysis. For instance, the combination of interviews, code reviews, and surveys helped Sharma et al. to enrich the overall findings of their study understanding security perceptions and practices of smart contract developers [69].

5.3 Target Population and Recruitment Challenges

Understanding the target population is crucial in human factors research, as it significantly influences the study's findings and applicability. In cybersecurity studies, researchers often focus on various groups, including:

General users: Many studies aim to understand the behaviors and perceptions of everyday users such as IoT users [87], university students/faculty [12, 18], Internet users [39], regarding cybersecurity practices. This group is essential for evaluating the usability and effectiveness of security measures, as their responses can reveal gaps in understanding and compliance.

Expert Professionals: Research targeting cybersecurity experts provides insights into the practices [3], challenges [19], and decision-making processes [44] of those responsible for maintaining security. This population often has a different perspective on risks and mitigations compared to general users.

However, recruiting experts is considered a challenging task within the broader security community due to their limited numbers, geographical concentration, and expectations for higher compensation [4]. Acar et al. offer initial evidence that GitHub users can serve as substitutes for experts in security tasks, providing an alternative recruitment avenue [4]. Additionally, recent work has explored the opportunities and challenges associated with various platforms for recruiting participants with software development experience for security-focused studies [40].

Vulnerable (at-risk) Populations: Some studies focus on specific demographic groups, such as older adults [29], journalists [53], sex workers [51], people with visual impairments [7], refugees [70], survivors of intimate partner abuse [9], content creators [64, 71], and other marginalized groups.

Understanding the unique challenges and perceptions of vulnerable (at-risk: those who experience heightened digital risks and harms [81]) population is essential for developing inclusive and effective security solutions since existing security, privacy, and safety threat models that tend to focus on a average user. And some attempt in that direction include the development of *at-risk framework* [81] and taxonomy of their online hate and harassment attacks [74]. These frameworks aim to highlight the specific vulnerabilities faced by marginalized populations, an attempt toward unifying language and frameworks to understand and address their needs more effectively.

5.4 Recruitment Strategies:

Common recruitment channels include flyers, physical message boards (on campus), personal contacts, university email lists, attendance at seminars/workshops, and online platforms such as Craigslist, Reddit, Facebook, Amazon Mechanical Turk (MTurk), and LinkedIn. A large number of cybersecurity studies use MTurk to recruit participants, but do the results generate approximate generalizability to the general population? To attempt that, Redmiles et al. compared users' self-reports about their security and privacy behavior, knowledge, past experiences, and advice sources across surveys conducted on MTurk ($n = 480$), using a nearly census-representative web-panel ($n = 428$), and via a probabilistic telephone survey ($n = 3,000$), finding that MTurk respondents tend to mirror the probabilistic sample – for those who are under the age of 50 [58].

Inclusion criteria are essential for ensuring that participants align with research objectives, enhancing data quality, improving study validity, optimizing resources, and maintaining ethical standards. A study understanding people's software updates encounters [47] reported this inclusion criteria:

> *We focused on finding adult Internet users that used Internet-enabled devices such as a desktop, laptop, tablet, or smartphone since they were likely to encounter software updates frequently.*

Many studies employ screening mechanisms to ensure the inclusion of the intended target group. For instance, a study on user mental models regarding privacy and security online [39] applied two different screening methods at separate stages: (a) educational level and college major as proxies for technical knowledge, and (b) a technical knowledge screening survey, where only participants scoring 5 or higher on an 8-item survey were considered technical participants.

5.5 Priming Techniques

Priming involves presenting information to participants that might influence their responses. Priming involves presenting information that might influence participants' responses. In cybersecurity studies, priming can focus participants on relevant issues [42, 48] but must be used cautiously to avoid introducing bias.

For example, when recruiting participants to understand their mental models for data breaches, study researchers reported that they:

> *recruited for a "study on personal finance and credit bureaus" purposefully not mentioning Equifax or identity theft to avoid priming participants and limit self-selection bias. [90].*

Whether to prime or not is a key consideration, as unintentional bias could affect the validity of the study's findings.

5.6 Ethical Considerations

Ethical concerns are addressed by adhering to Institutional Review Board (IRB) guidelines. This involves obtaining informed consent, ensuring participant confidentiality, and minimizing any potential risks. Researchers are required to transparently outline how data will be collected, stored, and used, and to implement mechanisms that allow participants to withdraw from the study at any time without consequences.

Ethical guidelines for research, particularly in studies involving human participants, emphasize minimizing potential psychological risks and respecting participants' well-being. For example, in a study with survivors of intimate partner abuse, the researchers made a deliberate choice to avoid asking questions that could lead participants to share traumatic or abusive experiences unrelated to the study's focus on technology [48].

5.7 Compensation

To encourage participation and acknowledge participants' time and contributions, compensation is generally offered. The type and amount of compensation can vary depending on the length and complexity of the study, as well as the availability of resources. Common forms of compensation include monetary incentives or gift cards.

5.8 Analysis Methods

Cybersecurity human factors research employs diverse qualitative analysis techniques [65]. Some of the common techniques are listed below.

Grounded Theory: Develops theories directly from data, making it ideal for exploring new phenomena.

Affinity Diagram: Visually organizes data into related groups, facilitating the identification of themes across user responses.

Open Coding Approach: Encourages flexible, unbiased exploration, allowing themes to emerge organically from the data.

Thematic Analysis: Involves identifying and analyzing patterns in the data, often employing inductive coding for data-driven insights. This method is particularly suited for exploratory studies.

Bottom-Up Inductive Coding: Starts with an initial list of themes, generating codes iteratively from the data to ensure findings are deeply rooted in participants' perspectives.

Generally, two or more researchers or coders are involved in the analysis process of the reviewed papers. This approach, known as double coding, aims to enhance rigor and transparency in the analysis. Various qualitative analysis tools, such as Dedoose, Atlas.ti, and MAXQDA, have been instrumental for researchers. Commonly reported inter-rater reliability measures include Cohen's Kappa, percent agreement, and Fleiss' Kappa. Additionally, researchers may employ Krippendorff's Alpha, which accommodates any number of raters and different types of data. These measures help ensure the reliability and validity of qualitative analyses by quantifying the consistency of interpretations among raters. The timing and rationale for employing inter-rater reliability are discussed in a meta-analysis by McDonald et al. [52].

6 Challenges and Limitations

In qualitative cybersecurity research, inherent challenges can affect the validity, reliabiality, and generalizability of findings. Issues with **recruiting participants**, including *social desirability* and *self-selection*, may influence how participants present themselves or opt into studies, potentially skewing results. Additionally, the reliance on **convenience sampling** and **small sample sizes** can further limit the representativeness of the data.

Given these concerns, it's crucial to recognize and address the constraints of qualitative research to bolster the reliability of insights drawn. This section provides a detailed examination of some common to "qualitative research" limitations, outlining the underlying reasons for each and suggesting strategies to mitigate their impact on the research process.

Sampling Size. Qualitative research often involves small sample sizes, which may not capture the full range of perspectives within a population. *Reasons:* Resource constraints and the intensive nature of data collection limit the number of participants.
Mitigation Strategies: Use *purposive sampling* to target a diverse range of participants, maximizing variability and covering a spectrum of experiences relevant to the research question.

Convenience Sampling. Can lead to sampling bias and limited generalizability, as participants are selected based on ease of access rather than representativeness. *Reasons:* Convenience sampling is often employed to save time and resources but may result in a homogenous sample that overrepresents certain demographics.
Mitigation Strategies: Combine with *stratified sampling* to ensure diversity and address representativeness, or conduct follow-up studies using alternative sampling methods to validate findings.

Self-Selection Bias. Self-selection can lead to an over-representation of certain traits, as well as social desirability bias, skewing results toward those with specific motivations or characteristics. *Reasons:* Individuals who volunteer may have strong opinions, a personal connection to the topic, or greater awareness, which could exclude perspectives from less-engaged or vulnerable groups.
Mitigation Strategies: Broaden recruitment efforts across multiple channels, consider offering *anonymity* to reduce social desirability bias, and compare self-selected sample data with more targeted recruitment efforts when feasible.

Qualitative Data Nature. Qualitative data can be subjective and difficult to quantify, making it harder to generalize findings and compare across studies. *Reasons:* The open-ended nature of data collection can lead to a range of responses that are highly context-dependent.
Mitigation Strategies: Employ *triangulation* by combining qualitative data with quantitative methods or using multiple qualitative methods to corroborate findings.

Recruitment Challenges Difficulty in recruiting a truly representative sample, especially for hard-to-reach populations, can limit the breadth of insights. *Reasons:* Participants may decline for reasons like privacy concerns, time constraints, or disinterest.
Mitigation Strategies: Use a mix of *incentives* to attract diverse participants, extend recruitment to multiple platforms, and conduct *pilot studies* to refine recruitment strategies.

Reporting Bias Participants may not fully disclose information or may alter responses to align with perceived researcher expectations (social desirability bias).
Reasons: Concerns about privacy, perceived judgment, or recall issues can affect the accuracy of responses.
Mitigation Strategies: Frame questions neutrally to avoid leading responses, assure *confidentiality*, and consider *double-coding* to enhance the validity of analysis by minimizing individual coder bias.

7 Conclusion: Integrating Human Factors into Cybersecurity

In conclusion, understanding human factors in cybersecurity is essential for advancing the Science of Security [34]. However, as a community, we must not rush into adopting qualitative approaches without a thorough understanding of their diverse methodologies and inherent challenges. Each qualitative method offers unique insights, but they also come with limitations that can impact our findings.

As the field continues to grow, it is imperative that we encourage more research focused on evaluating qualitative research methods to ensure that our collective understanding evolves with rigor and precision. It also ties in the importance of meta-analysis and peer review evaluation [72], which reflects on the progression and self-awareness of the field as it grows. By engaging in such reflective practices, we can better understand the nuances of our research and contribute to a more robust body of knowledge in cybersecurity. Future efforts should prioritize these evaluations to ensure that our collective work effectively addresses the complexities of human factors in this dynamic domain.

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